

Observable Compliance, Hidden Distortions: Cropland Protection under Incomplete Monitoring

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When Policy Enforcement Is Decentralized

- ▶ In a centralized system, the **central government** delegates development and policy targets to **lower-level governments**.
- ▶ **Multi-task bureaucrats** face **career incentives** to meet multiple targets.
- ▶ With **asymmetric information**, local governments may deviate from central objectives.

⇒ The center **strengthens monitoring** of previously neglected tasks.

⇒ When **monitoring is incomplete**, tighter supervision may **distort** behavior rather than correct it.

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- ▶ When the center tightens supervision on what it can observe, does it induce **within-task distortion** on what it cannot observe?

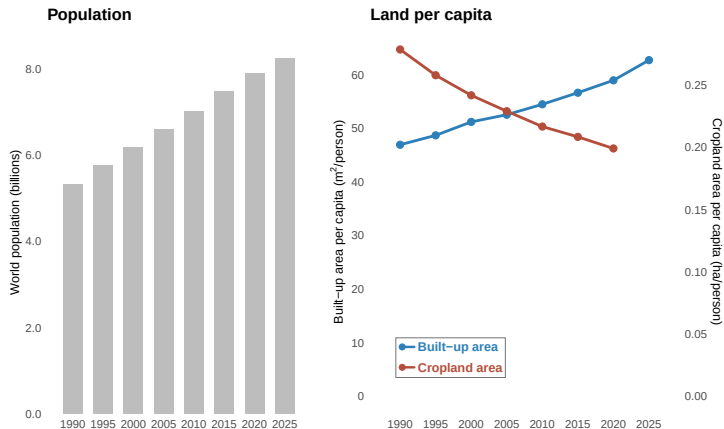
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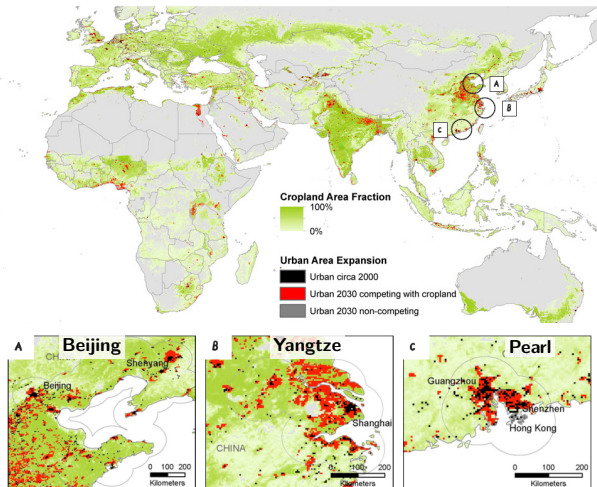
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The Land-Use Conflict Behind Urbanization and Population Growth



Sources: United Nations World Urbanization Prospects 2025 F01/F10; FAO 2025 FAOSTAT Land Use (accessed Jun. 2025, CC-BY-4.0). Cropland stops at 2020 because 2025 is not observed in the Jun. 2025 FAOSTAT release.

Cropland loss from urban expansion



Source: Adapted from Bren d'Amour et al. (2017), PNAS, Figs. 1–2. Red indicates projected urban expansion on cropland by 2030. Panels: A = Beijing-Tianjin-Hebei; B = Yangtze River Delta; C = Pearl River Delta.

Estimated Cropland and Crop Production Losses from Urban Expansion

Global estimates

Outcome	2000–2030	2015–2100
Cropland loss	27–35 Mha ¹	22.0–37.6 Mha ²
Crop production loss	3.4–4.2% ¹	1–4% ²
New urban land from cropland	–	51–63% ²

China estimates

Outcome	2000–2030
Cropland loss	7.1–8.6 Mha ¹
Crop production loss	8.2–9.8% ¹

Governments respond with different policy instruments.

¹ Bren d'Amour et al. (2017, *PNAS*); ² Chen et al. (2020, *Nature Communications*).

China's Policy Response: Cropland Occupation–Compensation Balance

Policy rule

When cropland is converted to construction land, local governments must compensate by creating an **equal amount** of cropland elsewhere.

Incentive tension

- ▶ **Multitask bureaucrats:** GDP (urban expansion) $\leftrightarrow$ cropland protection.
- ▶ **Within-task distortion** on cropland protection: quantity (monitoring strengthened after 2017) vs. quality (monitoring not strengthened).

Research Overview

- ▶ **RQ:** When cropland monitoring becomes more stringent on *area* but not *quality*, does local implementation distort land allocation?
- ▶ **Background**
 - ▶ 2017 policy tightening: stricter enforcement of cropland area targets.
 - ▶ Jiangsu: high conflict between agricultural responsibility and economic-development pressure.
- ▶ **Data**
 - ▶ Satellite imagery + Admin data.
 - ▶ Unit of analysis: 500 × 500 m grid.
- ▶ **Method**
 - ▶ **DiD:** high- versus low-pressure counties before/after 2017.
 - ▶ **Spatial RD-DID:** nearby grid cells across administrative borders.
 - ▶ Pressure = predicted compensation burden on a county's rural land reservoir.
- ▶ **Results**
 - ▶ Quantity ↑, but new cropland is **less suitable** and **used less** (vegetation ↓, fallow ↑).
 - ▶ Quality losses go beyond exhausting good land: **distorted reclamation choices** toward **less-monitored** and **lower-cost** sites; larger losses under stronger **career incentives**.

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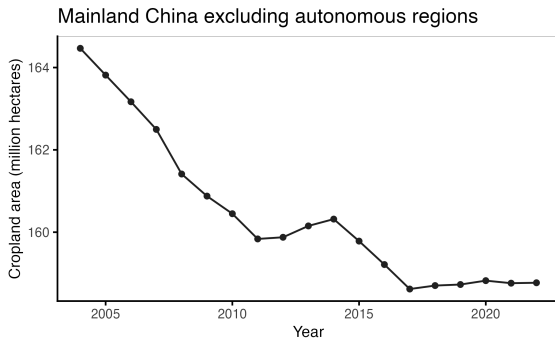
Related Literature

- ▶ **Principal-Agent Problem.** Holmstrom and Milgrom (1991); Baker (1992); Dixit (2002); Courty and Marschke (2004); Bandiera et al. (2021); Chen et al. (2018); He et al. (2020); Cao et al. (2026); Hong and Huang (2025); Fan et al. (2025); Fang et al. (2023); Axbard and Deng (2024); Fisman et al. (2025).
- ▶ **Decentralization and Local Incentives.** Montinola et al. (1995); Jin et al. (2005); Zhang (2006); Xu (2011); Li and Zhou (2005); Zhang et al. (2018); Wang et al. (2020); Wang and Yang (2025); Luo et al. (2025).
- ▶ **Land Regulation in China.** Cai et al. (2017); Henderson et al. (2022); Fu et al. (2021); Kong et al. (2024); Fang et al. (2026); Lu et al. (2026); Hong et al. (2025); Chen and Li (2025); Xing et al. (2026); Yu (2026).
- ▶ **Land Misallocation and Agricultural Productivity.** Restuccia and Santaaulalia-Llopis (2017); Adamopoulos and Restuccia (2014); Adamopoulos et al. (2022); Chari et al. (2021); Chen et al. (2022); Chen et al. (2023); Adamopoulos et al. (2024).
- ▶ **Contribution:** First evidence on within-task distortion under incomplete monitoring

Background: Cropland Occupation–Compensation Balance

- ▶ Long-standing policy^a – strengthened in 2017
- ▶ Core rule: occupy one unit, compensate one unit
- ▶ Implementation:
Upper-level government → lower-level government
assigns **planned land use quota (cap)**
- ▶ **Planned land use quota:**
new construction land converted from agricultural land
= cropland-compensation obligation

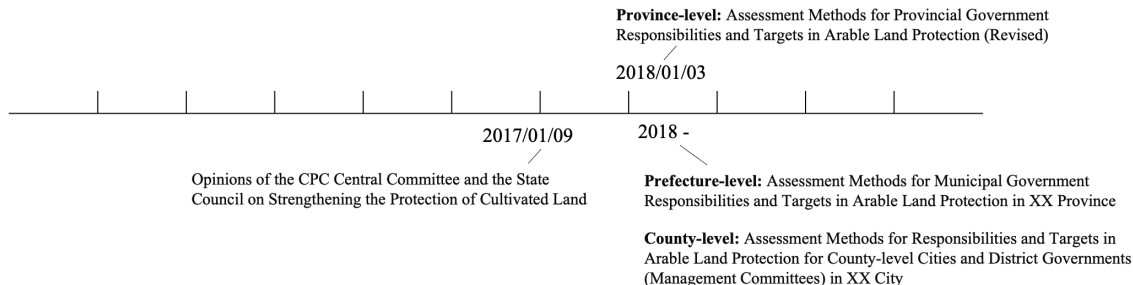
^a(in Chinese: 耕地占补平衡政策)



Full mainland sample

2017–2018 Policy Tightening

- **Cropland Occupation–Compensation Balance:** 2017–2018 revisions assessed local cropland protection more **stringently**.¹



¹Document of 2018 policy

Background: Land Regulation and Local Implementation

- ▶ **Central government's objective:** quantity-based and productive-capacity-centered
- ▶ **Documented Issues**
 - ▶ Implementation Distortion: ✓ quantity, ✗ quality (People's Daily)
 - ▶ Incentive design flaws and inadequate policy coordination (Ministry of Agriculture and Rural Affairs)

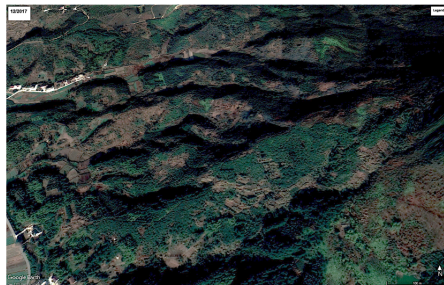


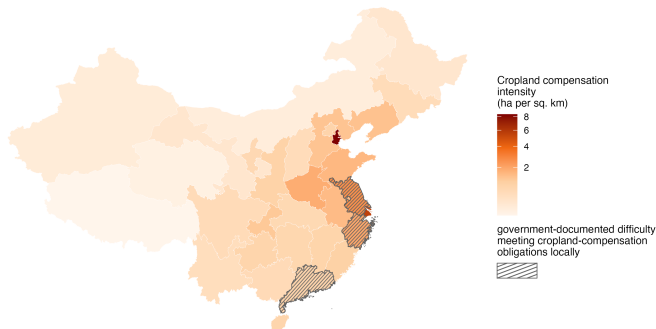
Figure: An example of local policy implementation: forest → terrace

Background: Regulation and Local Implementation

- ▶ **Multitask bureaucracy:** [economic growth](#) + [cropland protection](#).
- ▶ **Land-use regulation: Government-led designation:** land → [construction use](#) / [agricultural use](#).
 - ▶ **Construction land:** designation → announcement → bidding → land-use-right conveyance.
 - ▶ **Cropland:** reclamation → agricultural designation → contracting / leasing / transfer → cultivation.

Why Jiangsu?

- ▶ **Cropland-Compensation Intensity** $_p = \text{obligation}_p \text{ (耕地补充义务量)} / \text{provincial land area}_p$
- ▶ **Not all provinces face a binding land-use conflict.**
- ▶ **Jiangsu is a sharp case:** a major agricultural and economic province, mostly flat.



Data

Variable	Proxy	Data
Remote-sensing and raster data		
Cropland	Land cover	CLCD, 30 m annual land cover
Cultivation intensity	Vegetation cover	FVC, 30 m / 15-day Fractional Vegetation Cover
Quality	Soil type	China Soil Map (1995) and HWSD v2 (2023)
Administrative and economic data		
Officials		Public biographies of prefecture- and province-level officials
Boundaries / controls		Mainland China county map; County Statistical Yearbooks
Planning obligations		Prefecture Land Use Master Plans (2006–2020)
Land transaction		Jiangsu Land Market transaction records
Growth pressure	GDP growth targets, Q1 GDP growth	Government Work Reports

Measurement of Key Outcomes

Observation unit: 500 m × 500 m grid cell (applies throughout).

1. **Cropland.** *Cropland share*: fraction of 30 m pixels classified as cropland in each grid.
2. **Quality.** Mapping soil types to five-score suitability classes (FAO 1976; FAO 2007). A *binary variable* equals 0 for saline–alkali or other non-arable land.
3. **Cultivation intensity.**
 - ▶ Annual Fractional Vegetation Cover (FVC); higher FVC indicates denser vegetation cover.
 - ▶ *Fallow*: annual average FVC falls below 0.20 (bottom 25% of the distribution).

Reclaimed cropland: a grid whose cropland share is low in year $t - 1$ and rises to a high enough level in both t and $t + 1$.

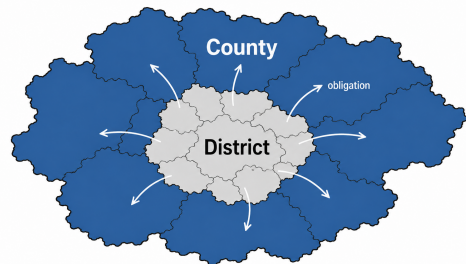
[Full data table](#)

Measurement of Cropland Protection Pressure

Cropland Occupation–Compensation Balance (耕地占补平衡政策)

- (1) Equal-area compensation (one-for-one).¹
- (2) Compensation is not necessarily local.
 - ▶ within the same county,
 - ▶ if infeasible (e.g., urban core district), within the same prefecture (across counties)².

This creates a *capacity constraint* determined by the non-core (non-urban) area within a prefecture.



¹占—补—: a county converting cropland to construction land compensates with an equal amount of cropland elsewhere.

²Jurisdictions that cannot meet the balance requirement can purchase supplementary cropland indicators through a paid, auction-based transaction platform.

Measuring County-Level Cropland Protection Pressure

Definition: Cropland Protection Pressure

$$\text{Pressure}_c = \left(\frac{\text{Obligation}_p}{\text{TotalLand}_p} \right) \times \text{LandShare}_c,$$

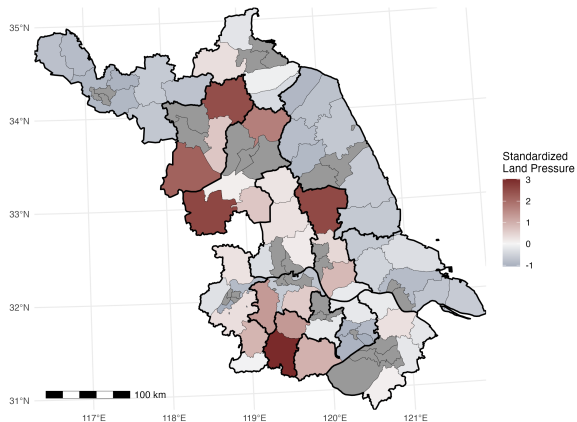
- ▶ Obligation_p : Total planned cropland-compensation obligation assigned to the *core urban districts*¹ of prefecture p under the provincial planning targets (set in 2017, effective 2018–2020).
- ▶ TotalLand_p : Total area of *available land*² in prefecture p outside the urban districts. This is the prefecture's “land reservoir” available for real cropland compensation.
- ▶ LandShare_c : County c 's share of this reservoir:

$$\text{LandShare}_c = \frac{\text{AvailableLand}_c}{\sum_{c' \in p} \text{AvailableLand}_{c'}}.$$

¹Specified in Land Use Master Plan. Fully core = 1, partly core 0.5, non-core = 0.

²1 minus pre-policy mean impervious share minus pre-policy mean water share.

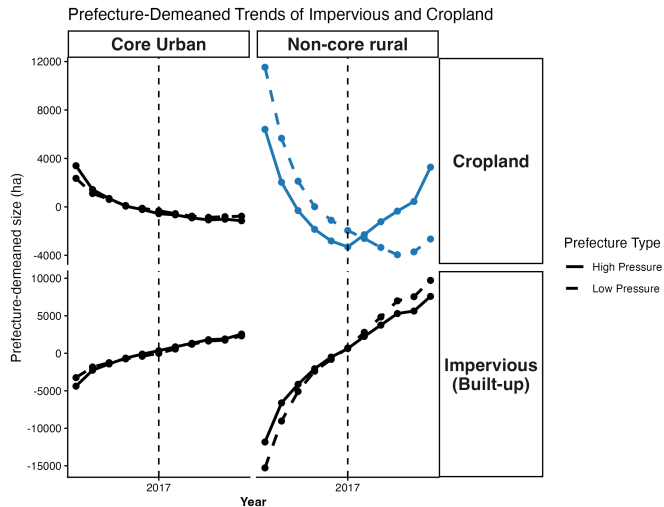
Standardized Land Protection Pressure in Jiangsu



Note: Positive values = above-average exposure; gray = core urban districts.

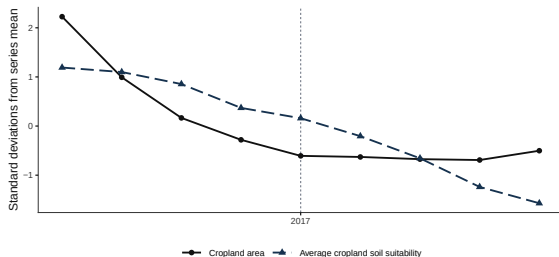
Pressure distribution

Motivating Fact

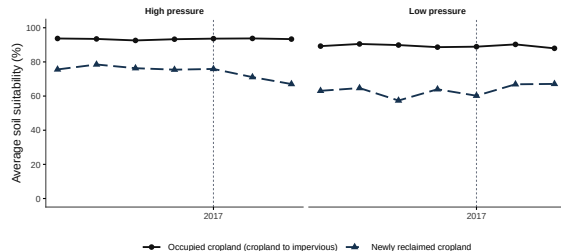


Motivating Facts: Cropland Quantity and Quality

Area and average suitability



Occupied vs. newly reclaimed land

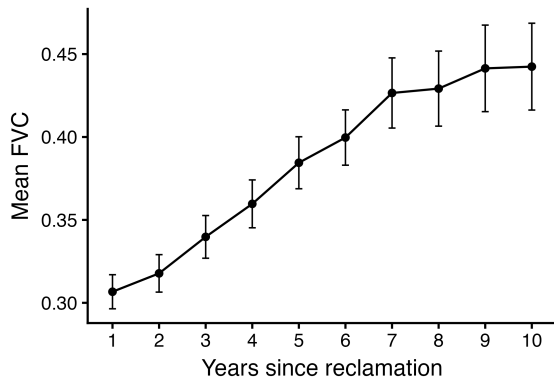


- ▶ After 2017, cropland area is broadly stable, while **average soil suitability continues to decline**.
- ▶ Occupied cropland remains more suitable than newly reclaimed land; reclaimed-land suitability **falls in high-pressure prefectures**.

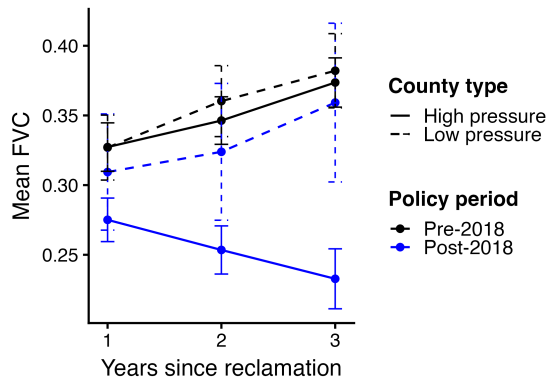
Notes. Occupied: cropland → impervious surface; newly reclaimed: noncropland → cropland. Left: standardized series; mean suitability among cropland-dominant grids. Right: suitable share (1995 soil score ≥ 3) of 30 m transition pixels, stable four years before and after conversion (through 2019). Pressure groups split at the prefecture median; dashed lines mark 2017.

Motivating Fact: Cultivation intensity and FVC

Mean FVC of Newly Reclaimed Land
in Jiangsu Province



Mean FVC by Land Pressure
and Policy Period



Identification Specification

(1) Event-study DID: cropland share of all grids

$$\text{CropShare}_{ict} = \sum_{k \neq -1} \beta_k (\text{Pressure}_c \times \text{Event}_{k,t}) + \mu_i + \lambda_t + \varepsilon_{ict}$$

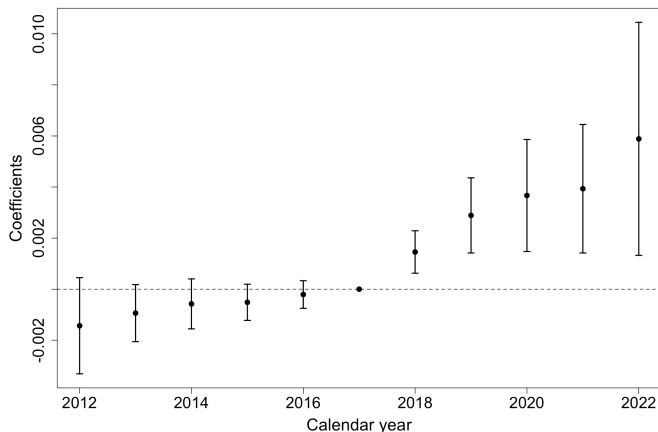
- ▶ $\text{Event}_{k,t}$: event-time indicators relative to the 2017 tightening ($k = -1$ omitted).
- ▶ **FE**: grid (μ_i) + year (λ_t).

(2) Cohort DID: fallow & soil suitability of newly reclaimed land

$$Y_{i,s} = \beta (\mathbf{1}\{t_0(i) \geq 2018\} \times \text{Pressure}_c) + \alpha_c + \mu_{t_0(i)} + \delta_s + \varepsilon_{i,s}$$

- ▶ The outcomes are attributes of the newly reclaimed land — fallow and soil suitability undefined before reclamation $\Rightarrow$ cohort DID.
- ▶ **Cohort** $t_0(i)$ = grid i 's reclamation year.
- ▶ **FE**: county (α_c) + cohort ($\mu_{t_0(i)}$) + years-since-reclamation (δ_s).

Cropland Share Expansion



- 1 s.d. $\text{Pressure}_c \uparrow \Rightarrow +0.16 \text{ p.p.}$ cropland share in 2018 ($\approx 13,700 \text{ ha}$); $\approx 84\%$ of the pre-period annual cropland expansion.

Relative to 2017; pre-period annual expansion = avg. gross cropland gain 2013–2017 in the non-core grid sample ($\approx 16,200 \text{ ha/yr}$).

Soil Suitability

	Soil suitability				1(Fallow)	
	China Soil Map 1995		HWSD v2 2023			
	(1)	(2)	(3)	(4)	(5)	(6)
Pressure $\times$ Post	-0.051 ^{**} (0.020)	-0.057 ^{***} (0.021)	-0.032 [*] (0.018)	-0.037 [*] (0.019)	0.055 ^{***} (0.015)	0.056 ^{***} (0.017)
Control Mean	0.593	0.593	0.573	0.573	0.198	0.200
County controls $\times$ year		✓		✓		✓
County & Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE					✓	✓
Cluster level	County	County	County	County	County	County
Number of counties	52	52	52	52	54	53
Num. Obs.	939	939	939	939	3,244	3,222
R ²	0.452	0.457	0.490	0.491	0.232	0.230

- 1 s.d. $\text{Pressure}_c \uparrow \Rightarrow \downarrow \mathbf{5.6 \text{ p.p.}}$ prob. suitable soil in 1995 ($\downarrow 9.4\%$ of baseline) and $\downarrow \mathbf{4.3 \text{ p.p.}}$ in 2023 ($\downarrow 7.5\%$).

Baselines: 0.596 (1995), 0.573 (2023).

Cultivation Intensity

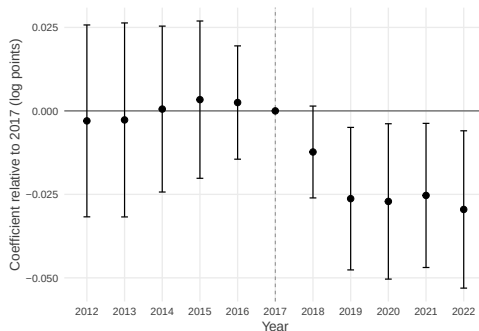
	(1)	(2)
Land protection pressure $\times$ post-2017	0.055*** (0.015)	0.055*** (0.017)
Control mean	0.198	0.198
County controls $\times$ year		✓
County and cohort FE	✓	✓
Years-since-reclamation FE	✓	✓
Cluster level	County	County
Number of counties	54	53
Num. Obs.	3,244	3,222
R ²	0.231	0.230

- Pressure_c $\uparrow$ after 2017 $\Rightarrow$ +**5.5 p.p.** fallow probability (+28% of baseline).

Baseline (control mean) = 0.198.

Fallow event-study figure

Cultivation Intensity: Crop Output per Cropland Hectare



- ▶ Before 2017, crop output per hectare shows **no differential trend** ($p = 0.741$).
- ▶ From 2019–2022, a 1 s.d. increase in pressure implies **2.5–3.0% lower output per hectare** ($p < 0.05$ each year).

County-year event study; grain + oilseed + cotton output per satellite-measured cropland hectare; county and year fixed effects; standard errors clustered by county.

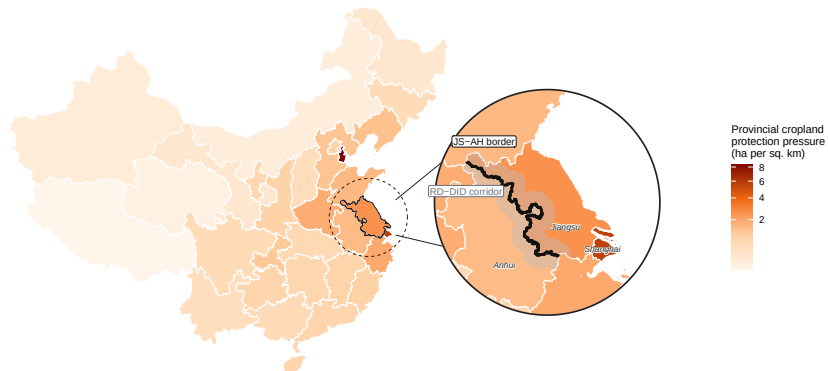
Robustness Checks

- ▶ **Parallel trends for the cultivation-intensity measure**
- ▶ **Alternative definitions of land pressure**
- ▶ **Sensitivity to reclaimed-land classification**
- ▶ **Sensitivity to fallow definition**

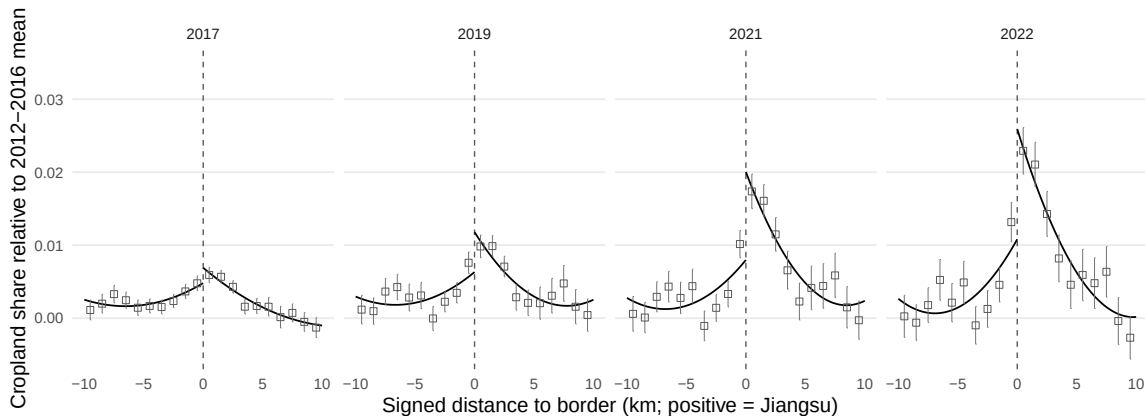
[Event study of FVC](#)[Appendix table](#)[Appendix figures](#)[Appendix figure](#)[◀ Back to specification](#)

Spatial RD-DID

Compare pre-post change in outcome across the border.

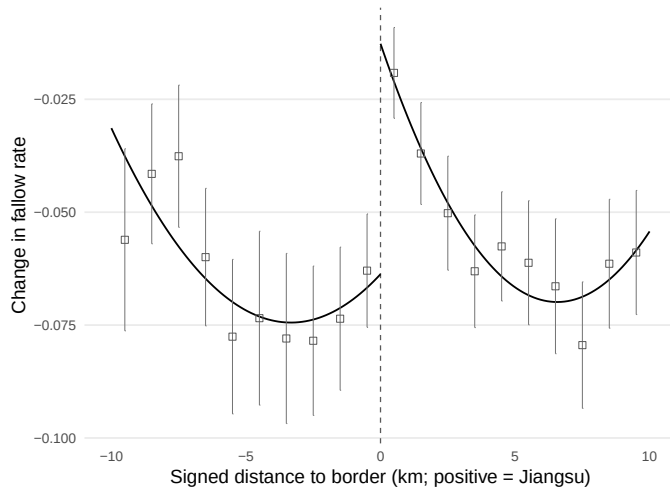


Spatial RD-DID: Δ Cropland Share (vs. 2012–2016 Mean) at the Jiangsu–Anhui Border



Placebo: assign policy to 2013/14 $\Rightarrow$ no pre-period border gap

Spatial RD-DID: Δ Fallow Rate at the Jiangsu–Anhui Border



Spatial RD-DID: Δ Cropland Share and Δ Fallow Rate

	Main comparison			Robustness comparison		
	Jiangsu–Anhui			Jiangsu–Shanghai		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Δ Cropland share</i>						
RD in Δ cropland share	0.006**	0.006**	0.007**	−0.027*	−0.027*	−0.029**
(Jiangsu – comparison side)	(0.003)	(0.003)	(0.003)	(0.014)	(0.015)	(0.012)
<i>Panel B: Δ Fallow rate</i>						
RD in Δ fallow rate	0.037***	0.038***	0.037***	−0.256***	−0.266***	−0.266***
(Jiangsu – comparison side)	(0.010)	(0.011)	(0.014)	(0.035)	(0.031)	(0.042)
County-pair indicators	Yes	Yes	Yes	Yes	Yes	Yes
Kernel	Triangle	Epanech.	Uniform	Triangle	Epanech.	Uniform

Notes. Outcomes are 2018–2022 means minus 2012–2017 means. Local-linear RD-DID in pre-to-post changes within the ± 50 km corridor, with additive county-pair indicators and MSE-optimal bandwidths. Entries are robust bias-corrected estimates; robust standard errors clustered by county are in parentheses. Panel B uses complete-FVC ex ante marginal cells.

Additional Results: Spatial Allocation and Environmental Outcomes

	Spatial allocation			Conversion to cropland		
	Dist. to existing cropland (m) (1)	Zero-cropland grids (2)	Isolated grids (3)	Water (4)	Forest (5)	Wetland (6)
Pressure × Post-2017	4.222** (1.782) [0.035]	0.470** (0.184) [0.032]	0.263** (0.123) [0.045]	0.082*** (0.018) [< 0.001]	-0.010 (0.008) [0.238]	0.038*** (0.009) [< 0.001]
<i>Magnitudes:</i>						
1-SD Pressure	4.662**	0.519**	0.291**	0.090***	-0.011	0.042***
Outcome mean	11.011	1.189	0.311	0.214	0.046	0.087
County FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Observations	640	640	640	576	576	576
R^2	0.336	0.324	0.200	0.690	0.478	0.587
County clusters	64	64	64	64	64	64
Pre-policy joint p -value	0.317	0.600	0.288	0.101	0.366	0.335

Notes. County-year regressions: 2013–2022 in cols. (1)–(3); 2013–2021 in cols. (4)–(6). Post: 2018 onward. SEs clustered by county in parentheses; BH-adjusted q -values in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Distance uses nearest prior-year cropland ($\geq 1\%$); isolated grids and their within-county rook neighbors have $< 1\%$ prior-year cropland. Cols. (2)–(3): shares of positive cropland gains; cols. (4)–(6): annual conversion shares. Col. (1) is in meters; other effects are in p.p. Water/forest: CLCD; wetland: GAWD; categories can overlap.

Result. More isolated new cropland; greater conversion of [water bodies and wetlands](#).

Alternative Explanations for Higher Fallow

Potential alternative explanations for higher fallow

1. ✗ Normal seasonality

⇒ Month-specific cohort DiD: **higher fallow persists in April–May and July–September.**

Evidence

2. ✗ Temporary land preparation

⇒ Fallow by years since reclamation (crop-pattern data): **no gap in years 1–2; gap in year 3.**

Evidence

3. ✗ Crop switching

⇒ Crop-frequency and crop-composition estimates: **no clear shift among detected cultivation.**

Evidence

Mechanism: Costly Compliance or Distorted Choices?

Costly compliance

Reclaim best land first; quality falls only because good sites run out (*land-quality frontier*).

Distorted choices

Depart from the best feasible path: pick **less monitored** or **lower-cost** sites.

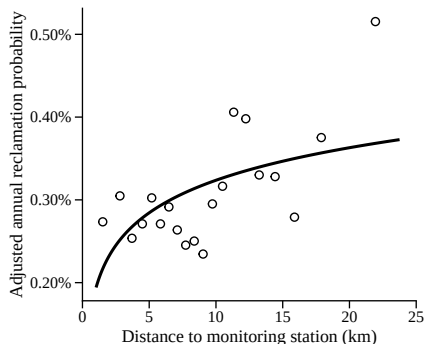
Evidence for distorted reclamation choices

1. **Avoiding quality monitoring**: after 2017, reclamation shifts *farther* from quality-monitoring stations, where new land is less suitable. [▶ Table](#) [▶ Figure](#) [▶ Coef. stability](#)
2. **Low-cost sites**: high-pressure counties reclaim more *previously cultivated* (cheaper to restore) land after 2017. [▶ Table](#)
3. **Frontier decomposition**: historical reference predicts quality from the remaining land stock; the **selection wedge explains 73–78%** of the suitability decline. [▶ Table](#)

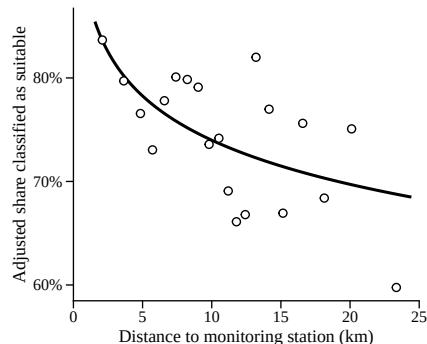
Incentives of local officials

- ▶ Suitability loss larger when the prefecture party secretary has **stronger promotion incentives** (triple interaction -0.058^{***}) [▶ Table](#); suggestive for GDP-growth pressure [▶ Table](#) [▶ Losses vs. land revenue](#)

Mechanism: Monitoring-Station Distance and Reclamation Choices



(a) Reclamation probability, 2018–22



(b) Suitable share of reclaimed land

- ▶ 100 m grids, pre-2017 provincial monitoring stations; farther from stations: **more reclamation, lower suitability**.
- ▶ Tightening *strengthens* the distance–reclamation gradient; quality gradient persists.

Mechanism: Selection of Previously Cultivated Land

- ▶ Prior cultivation proxies for lower reclamation costs.
- ▶ Higher-pressure counties shift toward previously cultivated land after 2017.

	100 × 1(Reclaimed)	
	(1)	(2)
	2 years	3 years
Historical-cropland definition		
Previously cultivated land × Post	−0.022 (0.032)	−0.024 (0.035)
Previously cultivated land × Post × Pressure	0.046** (0.017)	0.050** (0.020)
<i>Magnitudes:</i>		
1-SD Pressure	0.051**	0.055**
Grid FE	✓	✓
County-by-year FE	✓	✓
Observations	385,390	374,218
R^2	0.222	0.220
County clusters	64	64
Pre-policy pressure-gradient p -value	0.241	0.301

Notes. At-risk 500 m grid-years, 2013–2021; coefficients in p.p. Previously cultivated: cropland for ≥ 2 (3) consecutive years in 2001–2011 and noncropland in 2012; comparison: noncropland throughout 2001–2012. Post: after 2017; Pressure: county cropland protection pressure. SEs clustered by county; ** $p < 0.05$.

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Mechanism: Frontier Component vs. Selection Wedge

Selection accounts for 73–78% of the suitability decline.

Reference predictors	Total loss	Frontier component	Selection wedge	Frontier share (%)	Selection share (%)
Remaining suitable share	10.156	2.413	7.743	23.8	76.2
Remaining suitable share + depletion	10.156	2.204	7.952	21.7	78.3
Remaining suitable share + depletion + reclamation intensity	10.156	2.718	7.438	26.8	73.2

- ▶ **Reference:** 2002–2017 relation between each cohort's quality and the county's remaining reservoir, applied to 2018–2021 with actual quantities held fixed.
- ▶ **Frontier component:** historical mean – reference (costly compliance).
- ▶ **Selection wedge:** reference – observed quality (proxy for distortion).

Notes. 92 county-cohort observations in 37 counties, reclaimed-area weighted; suitable-land share (p.p.) under the 1995 soil map. Back-of-the-envelope decomposition.

[◀ Back to mechanism](#)

Mechanism: Career Incentives of Prefecture Party Secretaries

	China Soil Map 1995 Suitable share (1)	HWSD v2 2023 Suitable share (2)
Pressure × Post	−0.097 ^{***} (0.021) [0.006]	−0.067 ^{**} (0.022) [0.089]
Incentive × Pressure × Post	−0.058 ^{***} (0.013) [0.022]	−0.067 ^{***} (0.021) [0.067]
Control mean	0.593	0.573
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & cohort FE	✓	✓
Number of counties	52	52
Observations	939	939
R^2	0.498	0.532
Prefecture clusters	13	13

Notes. Suitable share: area-weighted source-pixel/component indicators for scores ≥ 3 . Post: after 2017. Stars use conventional SEs:

*10%, **5%, ***1%.

- **Promotion incentive** = 1 if the prefecture party secretary is ≤ 57 and the year precedes the 2020 national land inspection.
- Controls: age, male, Jiangsu birthplace.
- Parentheses: SEs clustered by prefecture; brackets: wild-cluster bootstrap p -values (9,999 draws).

Can Marketizing Obligation Reduce Land Misallocation?

- ▶ Jiangsu: online cropland quota platform available ✓; limited practical effect.
- ▶ Marketized **quota prices are far above reclamation fees**:
 - ▶ Local self-balance: 30–50 RMB/m²; cross-city quota transfer: 345–420 RMB/m²
- ▶ **Why high quota price?** High-pressure counties bear larger cropland-protection obligations and show **weaker post-2017 urban expansion**. ↘

	Total transacted construction-land area (1)	New construction- land supply (2)	Cropland-origin new construction land (3)
Pressure × post-2017	−0.150 ^{**} (0.058)	−0.180 ^{**} (0.085)	−0.154 [*] (0.091)
County, year FE & county trends	✓	✓	✓
Observations	704	704	640

Notes. **Magnitude:** 1-SD pressure ⇒ expansion down 12–16%.

- ▶ **Possible reason: equity–efficiency concern.** High trading costs may create misallocation but reduce spatial inequality across prefectures / provinces.

Wrap-Up

- ▶ **Policy:** 2017 tightening of China's **cropland occupation–compensation balance** policy
 - ▶ construction land conversion $\Rightarrow$ equal cropland compensation
 - ▶ binding area target $\uparrow$; quality monitoring remains weak
- ▶ **Main takeaway:** **single-task distortion in cropland protection**
 - ▶ weak quality monitoring + hard area target
 - ▶ $\Rightarrow$ quantity compliance $>$ productivity-preserving compliance
- ▶ **Mechanism:** central–local incentive misalignment
- ▶ **Policy implication:** marketizing cropland-protection obligations
 - ▶ local pressure $\downarrow$ $\leftrightarrow$ **equity–efficiency trade-off**

Thank You!



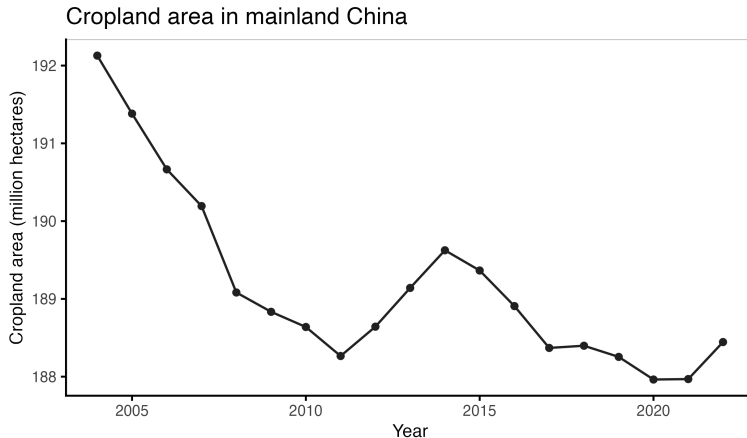
SWET slides (August 2026)



Paper (July 2026)

Ze (Emily) Wang · University of Tokyo

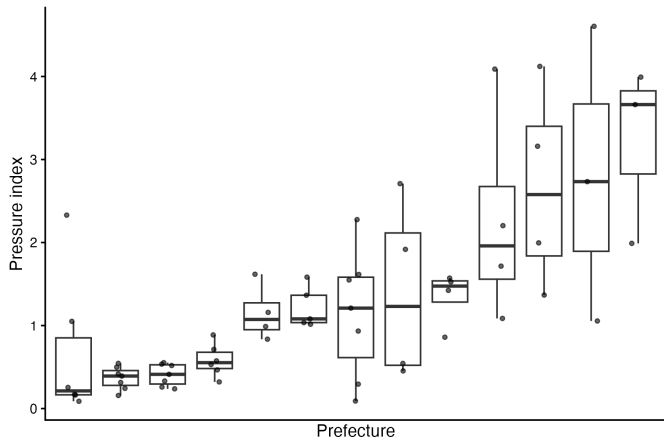
Full Mainland Cropland Area



Data source: CLCD county data. Mainland excludes Hong Kong, Macao, and Taiwan.

[◀ Back to main figure](#)

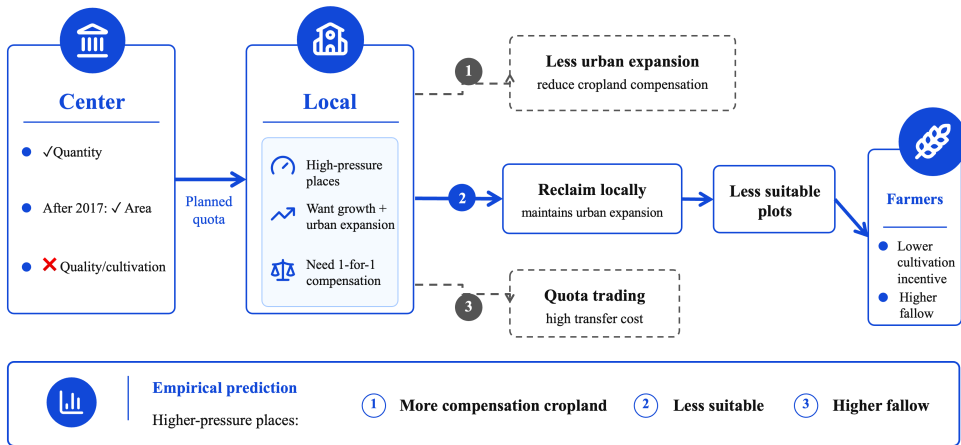
Distribution of Cropland Protection Pressure within and across Prefectures



Notes. The unstandardized pressure index is shown for non-core counties. Boxes indicate the interquartile range within each prefecture, thick lines indicate medians, and dots indicate individual counties.

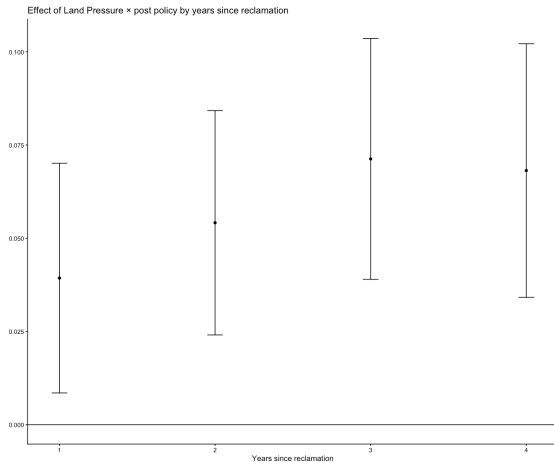
[◀ Back to map](#)

Conceptual Framework: From Imperfect Monitoring to Misallocation



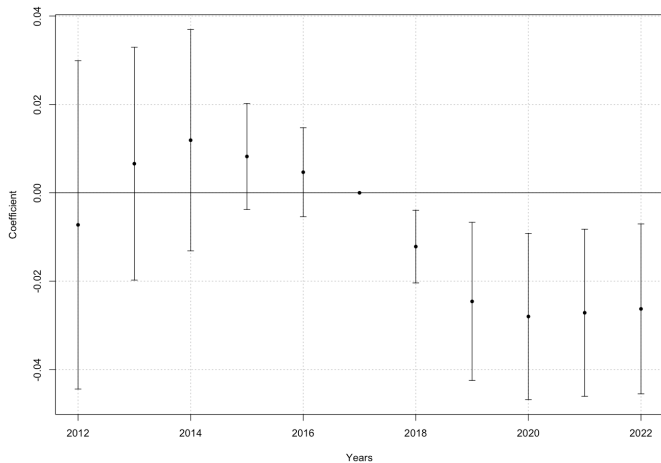
◀ Back to identification

Cultivation Intensity: Fallow Event Study



[◀ Back to cultivation intensity](#)

Event Study of FVC



[◀ Back to robustness check](#)

Alternative Definition of Land Pressure

Table: Robustness: Alternative Definitions of Land-Pressure Exposure

	(1) Cropland Share	(2) Soil suitability	(3) Fallow
Panel A: All Land Types as Available Land			
Pressure $\times$ Post	0.006*** (0.002)	-0.071*** (0.024)	0.078*** (0.026)
Grid & Year FE	✓		
County & Cohort FE		✓	✓
Years-since-reclamation FE			✓
Cluster level	County	County	County
Number of counties	64	52	54
Num. Obs.	3,743,982	939	3,243
R ²	0.981	0.452	0.230
Panel B: Alternative Definition of Core-Urban Districts			
Pressure $\times$ Post	0.003** (0.002)	-0.051* (0.028)	0.084** (0.032)
Grid & Year FE	✓		
County & Cohort FE		✓	✓
Years-since-reclamation FE			✓
Cluster level	County	County	County
Number of counties	64	52	54
Num. Obs.	3,743,982	939	3,243
R ²	0.981	0.451	0.234

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Sensitivity Analysis

► Sensitivity to Reclaimed-Land Classification

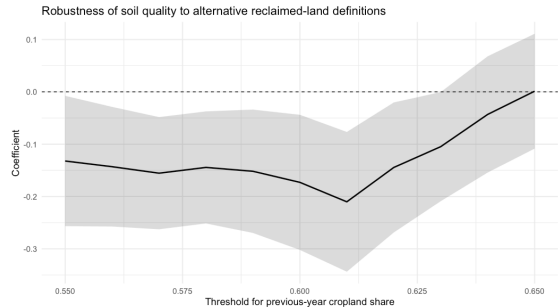
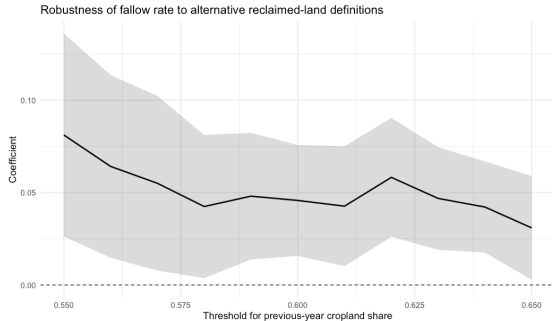
- Baseline: $\text{Reclaimed}_t = 1$ if $\{\text{CropShare}_{t-1} < 0.60\} \wedge \{\text{CropShare}_t, \text{CropShare}_{t+1} > 0.70\}$, where 0.70 is the pre-treatment mean.
- Robustness: 0.60 $\rightarrow$ 0.55–0.65 for the previous year's threshold.

► Sensitivity to Fallow Definition

- Baseline: Fallow = 1 if $\text{FVC} < 0.20$, where 0.20 corresponds to the bottom 25%.
- Robustness: 0.20 $\rightarrow$ 0.13–0.20.

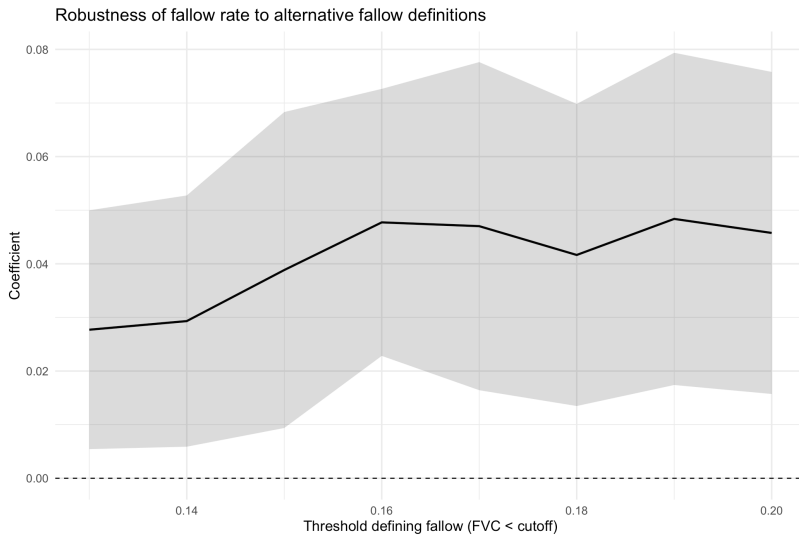
◀ [Back to robustness check](#)

Sensitivity to Alternative Thresholds in Reclaimed-Land Definition



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Sensitivity to Alternative Fallow Rate Definition



[◀ Back to robustness check](#)

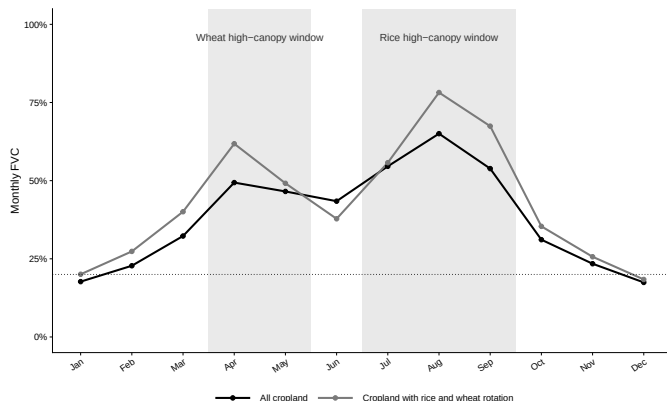
Placebo Test

	(1) Triangle	(2) Epanech.	(3) Uniform
<i>Panel A: Δ Cropland share</i>			
RD in 2013–2012	–0.0016	–0.0016	–0.0016
(Jiangsu – Anhui)	(0.0027)	(0.0026)	(0.0026)
RD in 2014–2012	–0.0007	–0.0005	–0.0004
(Jiangsu – Anhui)	(0.0034)	(0.0033)	(0.0033)
<i>Panel B: Δ Fallow rate</i>			
RD in 2013–2012	0.0112	0.0097	0.0073
(Jiangsu – Anhui)	(0.0093)	(0.0100)	(0.0100)
RD in 2014–2012	0.0018	–0.0005	–0.0086
(Jiangsu – Anhui)	(0.0157)	(0.0158)	(0.0161)

⇒ **No pre-policy discontinuity at the border.**

Notes. Placebo RD-DID assigning the policy to pre-policy years (2013, 2014); the outcome is the Δ cropland share (or fallow rate) relative to 2012. No pre-policy estimate is statistically significant. Full pre-period table (2013–2016) in the appendix.

Normal Seasonality: Cultivation Calendar

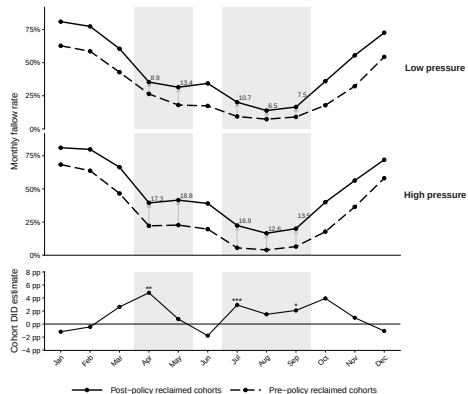


Finding. April–May and July–September are the high-canopy growing windows in both all cropland and rice–wheat rotation fields.

[Cohort-DiD estimates](#)

[◀ Back to alternative explanations](#)

Normal Seasonality: Month-Specific Fallow Estimates

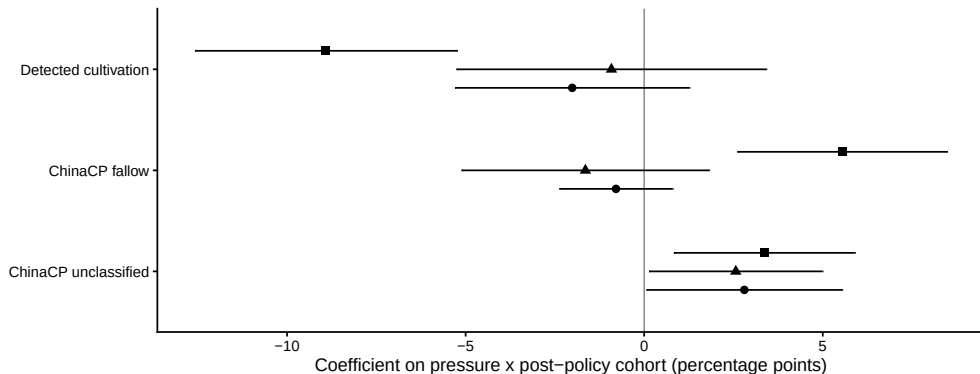


Finding. In high-pressure counties, the cohort-DiD estimates are positive throughout the April–May and July–September growing windows.

[Cultivation calendar](#)

[Back to alternative explanations](#)

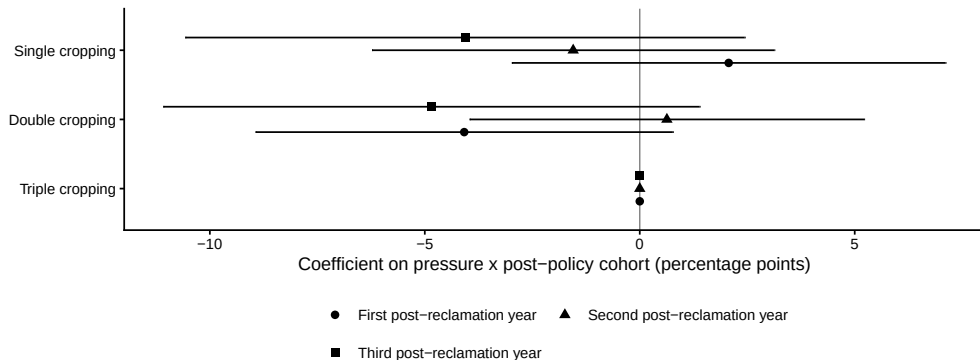
Land Preparation: Appendix Evidence



Finding. If preparation drove the result, the fallow gap would peak immediately after reclamation. Instead, there is no clear difference in years 1–2; in year 3, detected cultivation falls by **8.9 p.p.** and ChinaCP fallow rises by **5.5 p.p.**

[◀ Back to alternative explanations](#)

Crop Switching: Appendix Evidence



Finding. Among detected cultivation, pressure does not clearly change single-, double-, or triple-cropping shares. Nor is any crop-specific estimate significant after false-discovery-rate adjustment.

[◀ Back to alternative explanations](#)

Mechanism: Monitoring-Station Distance and Reclamation

	Extensive margin				Intensive margin: Suitable share			
	1(Reclaimed in year)				1995 China Soil Map		2023 HWSD v2	
	Level gradient		Grid FE					
	Baseline (1)	+ Urban (2)	Baseline (3)	+ Urban (4)	Baseline (5)	+ Urban (6)	Baseline (7)	+ Urban (8)
$\log(d_i) \times \text{Post}$	0.080*** (0.007)	0.066*** (0.007)	0.080*** (0.007)	0.066*** (0.007)	0.010 (0.014)	0.011 (0.014)	−0.004 (0.016)	−0.006 (0.016)
$\log(d_i)$	0.011** (0.004)	0.014*** (0.004)			−0.064*** (0.013)	−0.067*** (0.013)	−0.135*** (0.015)	−0.130*** (0.015)
Grid FE			✓	✓				
Year FE	✓	✓	✓	✓				
County FE	✓	✓			✓	✓	✓	✓
Reclamation-cohort FE					✓	✓	✓	✓
Baseline land-use controls	✓	✓			✓	✓	✓	✓
Urban-center distance control		✓		✓		✓		✓
Mean dependent variable	0.002	0.002	0.002	0.002	0.742	0.742	0.623	0.623
Observations	15,512,250	15,512,250	15,512,250	15,512,250	30,891	30,891	30,892	30,892
R^2	0.007	0.007	0.101	0.101	0.345	0.346	0.407	0.408
Cluster level				Parent 500 m grid				

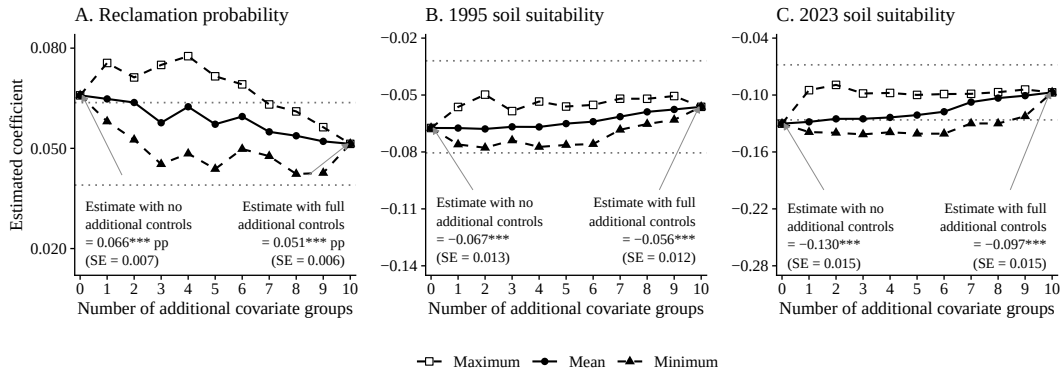
Notes. d_i : distance from 100 m grid i to the nearest pre-2017 provincial cropland-quality monitoring station. Cols. (1)–(4): grid-years 2013–2022, Post = 2018–2022, coefficients in p.p. Cols. (5)–(8): grids first reclaimed in 2013–2022. Soil outcomes: area-weighted shares of source pixels/components with score ≥ 3 , inherited from parent 500 m grids. SEs clustered by parent grid; * 10%, ** 5%, *** 1%.

► Coef. stability

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◄ Back to figure

Mechanism: Coefficient Stability of Monitoring-Distance Estimates



- ▶ Add subsets of **ten covariate groups**: county-boundary and government-seat distance; soil suitability, cropland, impervious, forest, and water shares within 5 km of the station; elevation; ruggedness; nighttime lights.
- ▶ (a) $\log(d_i) \times \text{Post}$ for annual reclamation (p.p.); (b)–(c) level coefficient on $\log(d_i)$ for 1995 China Soil Map / HWSD v2 suitable shares. Lines: mean, max, min across subsets; dotted: 95% CI of the ten-group specification.
- ▶ **Signs are retained across all combinations; fully adjusted estimates remain significant.**

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Mechanism: Economic Growth Pressure

- **Growth burden:** target growth minus predicted annual growth based on quarter 1; high burden when positive and in the top tercile in Jiangsu that year.

	Cropland share (1)	New construction land area (ha) (2)	Cropland-origin construction area (ha) (3)
Pressure $\times$ Post $\times$ growth burden	0.006* (0.003) [0.358]	267.707** (95.222) [0.340]	261.115** (86.557) [0.301]
Party-secretary controls	✓	✓	✓
Grid FE	✓		
Prefecture FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	3,050,309	116	116
R^2	0.985	0.604	0.637
Prefecture clusters	13	13	13

Notes. Pressure: prefecture cropland protection pressure; Post: after 2017. Col. (1): non-core rural cropland; cols. (2)–(3): urban-core conveyances signed after March 31. Cropland origin uses prior-year cropland at the transaction centroid. All lower-order interactions included. Parentheses: prefecture-clustered SEs; brackets: wild-cluster bootstrap p -values (9,999 draws). Stars use conventional SEs: *10%, **5%, ***1%.

Agricultural Losses and Local Incentives

- ▶ A production function maps suitability and fallowing effects into agricultural value-added losses.
- ▶ A one-standard-deviation increase in pressure implies an **11.1% annual loss**.

	Agricultural baseline	Primary-industry baseline
Annual loss (RMB/ha/year)	3,456	5,530
Capitalized perpetual loss (RMB/ha)	59,583	95,346
Share of conveyance revenue	0.14%	0.23%

- ▶ Nearby new construction-land conveyances yield **RMB 42.1 million/ha** in gross revenue.

Notes. All values in 2017 RMB. Capitalization holds annual losses constant. Revenue is total deflated payments divided by area for new construction-land conveyances within 5 km of land reclaimed in the same year.

[◀ Officials' incentives](#)[◀ Back to mechanism](#)

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